Manifold-Guided Adversarial Attack of Deepfake Detectors via Dual-Branch Autoencoder Projection

Stefano Cirillo
University of Salerno
Italy
scirillo@unisa.it

Luigi Di Biasi
University of Salerno
Italy
ldibiasi@unisa.it

Giuseppe Genito
University of Salerno
Italy
ggenito@unisa.it

Antonio Landi
University of Salerno
Italy
a.landi128@studenti.unisa.it

Mattia Maucioni
University of Salerno
Italy
m.maucioni@studenti.unisa.it

Dorotea Serrelli
University of Salerno
Italy
d.serrelli1@studenti.unisa.it

Giandomenico Solimando
University of Salerno
Italy
gsolimando@unisa.it

Abstract

We present DÆLTA, a Dual-branch AutoEncoder Latent Transformation Attack framework designed to generate adversarial versions of deepfake images in a transfer black-box setting, avoiding the need to optimize perturbations against a specific detector. DÆLTA projects each input through an autoencoder trained exclusively on a curated subset of real Flickr-Faces-HQ images. The architecture behind DÆLTA encodes each image via two parallel branches: one operating on RGB pixels and the other on log-scaled RGB Discrete Cosine Transform (DCT) representations augmented with self-attention. The resulting 256-dimensional latent feature vectors are concatenated and decoded by a shared transposed-convolutional decoder. Training is designed to minimize a composite perceptual loss combining different quality metrics following the guidelines of the ACM Multimedia Fake Face Detection Challenge (AADD 2026), on whose benchmark we evaluate DÆLTA against 2 white-box and 8 black-box detectors with heterogeneous architectures. Our method demonstrates cross-model transferability, with an Attack Success Rate (ASR) above 86% on six out of eight black-box detectors, outperforming many gradient-based baselines. To complement detector and perceptual measurements, we further introduce a supplementary, reproducible vision-language audit that compares captions and five controlled visual attributes of the original and attacked images. On the 1,600-pair collection, the audit reports BERTScore F1 of 0.9507 and Semantic Consistency of 0.6062 over 1,057 eligible pairs, showing that caption similarity alone can conceal changes in pose and background interpretation.

CCS Concepts

• **Computing methodologies** → **Computer graphics**; **Neural networks**; **Image representations**; • **Security and privacy** → *Biometrics*; *Intrusion/anomaly detection and malware mitigation.*

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Keywords

Deepfake Detection, Adversarial Attacks, Transferability, Autoencoders, Vision-Language Models, Semantic Evaluation

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1 Introduction

Recent advances in face synthesis have made AI-generated facial images increasingly realistic, challenging the reliability of forensic detectors that distinguish real from synthetic content.

Deepfake detection methods learn discriminative traces left by generative models, exploiting either spatial artifacts, such as texture inconsistencies and blending errors, or frequency-domain cues, including abnormal spectral patterns and Discrete Cosine Transform (DCT) irregularities [24]. Nevertheless, decision boundaries remain vulnerable to transformations that preserve facial semantics while altering the underlying feature representations.

Motivations and Proposal. While transfer-based black-box attacks have advanced the field of adversarial robustness, they remain reliant on surrogate classifiers, gradient approximations, or iterative query-based optimization, each introducing significant computational overhead and limiting practical feasibility. In this work, we investigate whether detector evasion can be achieved solely through manifold projection, without gradients, surrogate models, or repeated feedback. This is relevant to practical black-box settings where the adversary has no knowledge of the target architecture but can assume forensic detectors rely on generic anomalies in spatial pixels and high-frequency spectral domains. To address this, we propose DÆLTA, a Dual-branch AutoEncoder Latent Transformation Attack framework centered on autoencoder-based projection as its primary driver. Unlike conventional methods that optimize additive perturbations, our approach reconstructs the synthetic input via a decoder trained exclusively on authentic real facial images. This single forward pass implicitly anchors the adversarial output onto the learned manifold of real facial characteristics, inherently suppressing discriminative artifacts without task-specific supervision. To ensure an uncontaminated target distribution, we train our model on a rigorously curated subset of Flickr-Faces-HQ (FFHQ) [11],

pre-filtered via an AlexNet-based screening process to eliminate low-quality samples and residual synthesis traces. DÆLTA uses a dual-branch autoencoder that jointly processes RGB pixels and log-scaled DCT coefficients, simultaneously regularizing spatial inconsistencies and attenuating spectral fingerprints [7, 24, 26].

The official AADD protocol measures attack efficacy and visual similarity, but does not specify whether recognizable semantic attributes remain stable after the attack. We therefore add a clearly separated post-hoc semantic audit. A vision-language model (VLM) independently annotates each original/attacked pair with a neutral caption and a fixed set of observable attributes; the resulting measurements are reported alongside, but never combined with, the official Challenge Score, ASR, SSIM, or LPIPS. This audit characterizes the perceptual consequences of DÆLTA rather than modifying the attack or claiming a new benchmark for image-text evaluation.

2 Related Work

Existing research on detecting AI-generated facial content can be categorized into three main directions: *i*) deepfake and facial forgery detection methodologies, *ii*) adversarial attacks that evade detection systems, and *iii*) manifold projection and autoencoder methods for modeling and detecting synthetic faces.

Forensic technology has evolved across the spatial and frequency domains. Spatial methods analyze local artifacts, texture inconsistencies, and semantic features [10], while frequency-based approaches exploit spectral patterns and statistical anomalies in transformed domains such as Discrete Cosine Transform (DCT) or Fourier coefficients [4, 12]. Modern detectors often adopt hybrid paradigms, like F3-Net [24], to fuse spatial cues with high-frequency details. However, these models frequently rely on dataset-specific artifacts, leading to severe performance degradation under distribution shifts.

Adversarial attacks exploit detector vulnerabilities via white or blackbox perturbations [3]. Fast Gradient Sign Method (FGSM) and Projected Gradient Descent (PGD) show limited transferability [20], whereas Momentum Iterative Fast Gradient Sign Method (MI-FGSM) [6] and ensemble attacks improve cross-model evasion through multi-model gradient aggregation [6, 27]. Nevertheless, pixel-space attacks remain highly sensitive to model selection and frequently degrade visual quality. Recent literature has shifted toward latent-space manipulation within the generative process. Frameworks like LTA [28] leverage the VAE decoder as an implicit image prior, while AEGIS [13] injects signals along Denoising Diffusion Implicit Model (DDIM) denoising trajectories. DeDiAttack [25] incorporates elastic deformations to ensure naturalness, and universal frameworks [22] optimize specific timesteps under vision-language guidance. However, these approaches are closely bound to the diffusion backbone, leading to unstable trade-offs between fidelity and efficacy. In this context, manifold projection has shifted from defensive purification to offensive deepfake attacks by projecting deepfake samples onto a strictly constrained subspace of authentic human faces [14, 35], but existing latent and manifold methods still trade off transferability and visual fidelity [15].

To address these limitations, DÆLTA investigates whether robust evasion can emerge from manifold projection alone, bypassing the need for gradients or surrogate models. Unlike prior work that relies on backbone-specific priors, our method uses a dual-branch autoencoder that jointly encodes RGB and log-scaled DCT representations with self-attention, decoded through a shared network optimized with a composite perceptual loss. By replacing decision-boundary manipulation with statistical alignment to real data, our approach constrains deepfakes to a real-face subspace, weakening black-box forensic classifiers while preserving visual fidelity.

Beyond pixel- and feature-level fidelity, recent vision-language evaluation work uses decomposed questions to test whether an image supports individual facts [9], while contextual embedding metrics such as BERTScore compare generated text beyond exact word overlap [37]. These methods address image-text faithfulness rather than paired original/attacked deepfake images. Accordingly, our VLM-based analysis is not presented as a direct state-of-the-art comparison: it is a lightweight, fixed-attribute adaptation that measures relative semantic preservation within each pair.

3 The DÆLTA Approach

DÆLTA has been designed under the hypothesis that a fake facial image can be rendered less detectable by projecting it through a reconstruction model trained exclusively on the statistical properties of real faces, including natural texture patterns, color distributions, and frequency-domain characteristics. In contrast to gradient-based adversarial attacks, the proposed approach does not involve optimizing the input image with respect to the target detector [27].

3.1 Real-face subset selection

We begin by constructing a reduced training subset from FFHQ (see Step 1 of Fig. 1). Since training on the full dataset would expose the autoencoder to an excessively broad and heterogeneous distribution of facial appearances, we restrict the training data to a more compact region of the feature space. Specifically, we randomly sample a reference image from FFHQ and designate it as a pivot image. All candidate images are then resized, intensity-normalized, and geometrically aligned using face detection, after which they are passed through AlexNet for encoding. We then compute the Euclidean (L2) distance between the resulting feature representations as follows: $d_i = \left\|\phi(x_i) - \phi(x_p)\right\|_2$, where x_i denotes a candidate FFHQ image, x_p is the randomly selected pivot image, $\phi(\cdot)$ is the AlexNet feature extractor, and d_i is the Euclidean (L2) distance between their feature representations. We then let τ_p denote the p-th percentile of the empirical distribution of $\{d_i\}$ computed over all candidate FFHQ images. The training set is then defined as: $\mathcal{D}_{\text{train}} = \left\{x_i \,\middle|\, d_i \le \tau_p\right\}$, where x_i denotes a candidate FFHQ image, x_p is the randomly selected pivot image, $\phi(\cdot)$ is the AlexNet feature extractor, and d_i is the L2 distance between their feature representations. The resulting images, i.e., those falling within the p-th percentile of the distribution, are used as the real-image training set. The goal of this step is not to replicate the pivot image's identity, but to constrain the variability of the modeled real-image distribution during training.

3.2 Dual-branch autoencoder

The proposed autoencoder architecture comprises two parallel encoding branches that process complementary representations of the input image, as illustrated in Step 2 of Fig. 1.

Pixel branch (**Pixel-Æ**). The first branch operates directly on the RGB pixel space. By processing the input in the image domain, this branch preserves appearance-related cues associated with the visual content of the face, including fine-grained texture variations and spatial relationships among facial regions. The branch encodes

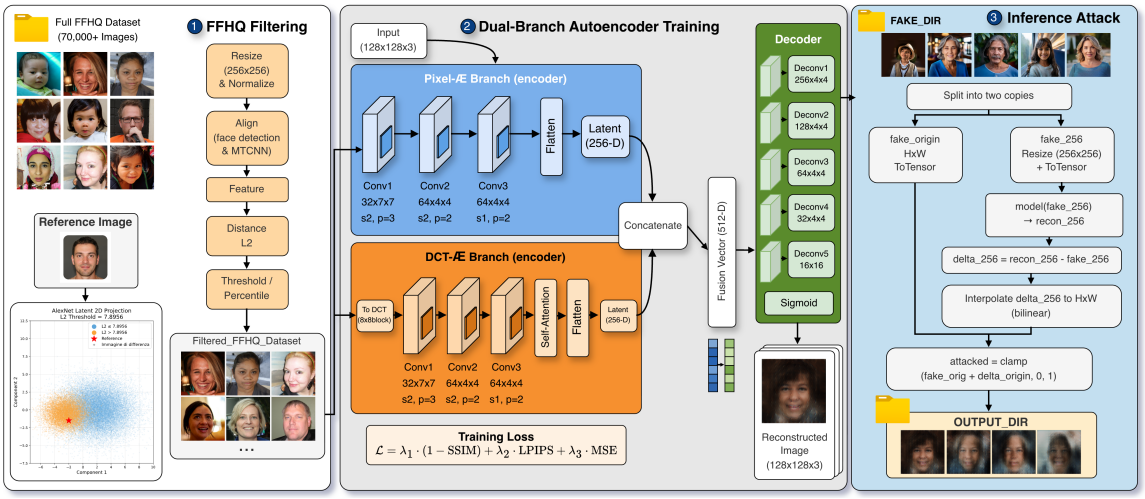


Figure 1: Overview of DÆLTA approach.

these characteristics using multiple convolutional layers, enabling the model to preserve fine-grained details and broader context [26].

Frequency branch (DCT-Æ). processes the log-scaled RGB DCT representation to model spectral regularities often less discernible spatially. The log scaling compresses the DCT dynamic range, balancing low- and high-frequency components [7, 17, 24]. A self-attention block is added before latent fusion to capture dependencies across distant or differently ranged spectral components, a strategy widely adopted in frequency-aware detection frameworks [4, 31]. Each branch independently maps its input to a 256-dimensional latent via convolutional layers, flattening, and linear projection. The resulting 512-dimensional concatenated vector is fed to a shared transposed-convolutional decoder, which reconstructs the RGB image at training resolution, effectively fusing complementary cross-domain information [33].

Training objective. To balance pixel-wise accuracy, structural consistency, and perceptual similarity, we minimize a composite loss that combines Mean Squared Error (MSE) [23], SSIM [32], and Learned Perceptual Image Patch Similarity (LPIPS) [36]: $\mathcal{L} = \lambda_1 \cdot (1 - \text{SSIM}) + \lambda_2 \cdot \text{LPIPS} + \lambda_3 \cdot \text{MSE}$, such that $\lambda_1 + \lambda_2 + \lambda_3 = 1$, where λ_1, λ_2, and λ_3 are scalar weights controlling each term's contribution. SSIM preserves global structure, LPIPS enforces perceptual similarity in a learned feature space, and MSE ensures pixel-level fidelity.

3.3 Inference attack

At inference time, given an input fake image x, we generate its adversarial counterpart through a single forward pass of the trained autoencoder, as illustrated in Step 3 of Fig. 1. The process consists of three main steps: resizing, reconstruction, and delta injection.

First, the input image x is resized to the spatial resolution expected by the autoencoder (x_{res}), while the original image is retained at its native resolution. This resizing step is necessary because the autoencoder expects fixed-resolution inputs, while the attack is applied to images of arbitrary resolution.

Second, the resized image x_{res} is passed through the dual-branch autoencoder, which produces a reconstructed version $A(x_{\text{res}})$. The reconstruction error, or delta, is computed in the autoencoder's resolution space as: $\Delta_{\text{res}} = A(x_{\text{res}}) - x_{\text{res}}$. This delta encodes modifications from the real-face manifold, capturing spatial and frequency-domain corrections from both branches.

Third, the delta is upsampled via bilinear interpolation to match the original image resolution, yielding Δ, which is added to the original image. The result is then clipped to the valid pixel range $[0, 1]$: $x_{\text{adv}} = \text{clip}(x + \Delta,\ 0,\ 1)$, where $\text{clip}(v, 0, 1) = \min(\max(v, 0), 1)$ constrains each pixel value to the valid range, ensuring that the adversarial image remains a valid RGB representation. This delta-based formulation preserves more of the original fake image content than directly replacing it with the autoencoder output $A(x)$, while still injecting the manifold projection effect learned from real faces. The presence of the DCT branch enhances the attack's ability to affect frequency-level cues, which is particularly effective against detectors relying on spectral artifacts.

4 Experimental Evaluations

This section evaluates DÆLTA's ability to evade detectors while preserving visual fidelity, assessing attack performance and perceptual fidelity across datasets, training setups, FFHQ subsets, and real-face distributions.

4.1 Experimental Settings

4.1.1 Datasets Statistics. The autoencoder is trained on real facial images from FFHQ [11], using the subset selection strategy described in Sec. 3.1 (see also Step 1 of Fig. 1). To assess how training set size affects attack performance, we evaluate FFHQ subsets of 5K, 10K, and 14K images, together with a full-data configuration. The most effective subset size and its impact on attack effectiveness and visual quality, is discussed in Sec. 4.2.2.

To evaluate the influence of the real-face source distribution, we train the autoencoder on different datasets while keeping the training set size fixed. Specifically, we consider FFHQ, CelebA-HQ, a curated 30K-image subset of CelebA [19], VGGFace2 [2], and their pooled combination, denoted as MIX. For VGGFace2, we use a reduced version consisting of 50K images instead of the full dataset. The same preprocessing and filtering procedure described in Sec. 3.1 is applied independently to each source, retaining the same number

Loss Weights λ_1	λ_2	λ_3	S	SSIM	LPIPS	ASR (%) ViT-B/16	DenseNet-DCT
0.30	**0.60**	**0.10**	**2003.4075**	$\mathbf{0.7513}_{\pm 0.0825}$	$\mathbf{0.3906}_{\pm 0.0790}$	**96.12**	**93.44**
0.60	0.30	0.10	1367.5344	$0.8113_{\pm 0.0725}$	$0.3138_{\pm 0.0807}$	77.38	37.69
0.33	0.33	0.34	1306.7492	$0.8083_{\pm 0.0733}$	$0.2995_{\pm 0.0765}$	69.81	39.12
0.20	0.20	0.60	1549.9130	$0.8086_{\pm 0.0731}$	$0.3023_{\pm 0.0768}$	80.00	49.63
0.20	0.70	0.10	1478.3805	$0.7835_{\pm 0.0805}$	$0.3089_{\pm 0.0750}$	65.88	60.94
0.20	0.10	0.70	1505.6460	$0.8129_{\pm 0.0718}$	$0.3150_{\pm 0.0809}$	82.13	44.63
0.15	0.75	0.10	1626.1472	$0.7766_{\pm 0.0825}$	$0.3112_{\pm 0.0754}$	68.63	71.13
0.10	0.80	0.10	1587.7210	$0.7644_{\pm 0.0843}$	$0.3240_{\pm 0.0748}$	68.75	69.69
0.50	0.50	0.00	1217.7671	$0.8081_{\pm 0.0734}$	$0.3087_{\pm 0.0812}$	65.44	37.00
0.00	0.90	0.10	979.4036	$0.5037_{\pm 0.0862}$	$0.7468_{\pm 0.0835}$	74.00	87.56
0.90	0.00	0.10	1192.1109	$0.8167_{\pm 0.0703}$	$0.3242_{\pm 0.0828}$	66.81	33.38
0.70	0.20	0.10	1212.1392	$0.8142_{\pm 0.0712}$	$0.3125_{\pm 0.0813}$	69.69	32.12

	|FFHQ|	S	SSIM	LPIPS	ASR (%) ViT-B/16	DenseNet-DCT
PIXEL-Æ	5K(7%)	1862.4515	$0.8005_{\pm 0.0731}$	$0.3493_{\pm 0.0815}$	95.56	65.50
	10K (14%)	1432.7774	$0.8257_{\pm 0.0692}$	$0.3040_{\pm 0.0779}$	83.62	34.44
	14K (20%)	1309.7758	$0.8362_{\pm 0.0662}$	$0.2803_{\pm 0.0720}$	78.44	27.12
	Full (100%)	816.0278	$0.8604_{\pm 0.0592}$	$0.2269_{\pm 0.0636}$	49.69	12.62
DCT-Æ	5K (7%)	1388.7069	$0.6317_{\pm 0.1031}$	$0.6374_{\pm 0.0959}$	95.00	79.69
	10K (14%)	1449.2099	$0.6164_{\pm 0.1023}$	$0.6358_{\pm 0.0898}$	92.38	91.88
	14K (20%)	1236.4501	$0.6379_{\pm 0.1012}$	$0.6129_{\pm 0.0957}$	82.81	68.38
	Full (100%)	1375.4415	$0.6408_{\pm 0.1028}$	$0.6193_{\pm 0.0929}$	88.88	79.31
DÆLTA	5K (7%)	1884.6270	$0.7706_{\pm 0.0794}$	$0.3622_{\pm 0.0813}$	94.06	74.50
	10K (14%)	1503.8083	$0.7943_{\pm 0.0765}$	$0.3143_{\pm 0.0762}$	76.12	51.81
	14K (20%)	**2003.4075**	$\mathbf{0.7513}_{\pm 0.0825}$	$\mathbf{0.3906}_{\pm 0.0790}$	**96.12**	**93.44**
	Full (100%)	934.7609	$0.8032_{\pm 0.0756}$	$0.2860_{\pm 0.0709}$	36.56	40.62

Table 1: Ablation studies on loss weighting strategy (left side) and autoencoder architecture on FFHQ subsets (right side).

of training samples for each configuration. The analysis of these source distributions is presented in Sec. 4.2.3.

For the MIX dataset, we report the distribution of the AlexNet feature-space distances used during filtering. The processed pool contains 148K images: 70K from FFHQ, 28K from CelebA-HQ, and 50K from VGGFace2. The 21st-percentile distance threshold from the pivot image retains 31,080 images, from which the final training subset is sampled. The retained MIX samples have mean L2 distance 8.91 ± 0.60 from the pivot, with median 9.04 and 5th–95th percentile range 7.75–9.63, indicating a compact real-face region after filtering. The retained pool is composed of 13,745 VGGFace2, 11,896 FFHQ, and 5,439 CelebA-HQ images, showing that the filtering criterion does not simply preserve the original dataset proportions.

4.1.2 Evasion metrics. In our evaluation, we consider the percentage of adversarial fake images misclassified as real by the target detector, i.e., Attack Success Rate (ASR). Moreover, we compute the Fake Detection Rate (FDR) before and after the attack, i.e., $\text{FDR}_{\text{clean}}$ and FDR_{adv}, and its differences $\Delta\text{FDR} = \text{FDR}_{\text{clean}} - \text{FDR}_{\text{adv}}$, as well as the detector's Area Under the ROC Curve (AUC) computed over the combined real/fake test set before and after the attack, i.e., $\text{AUC}_{\text{clean}}$ and AUC_{adv}, with $\Delta\text{AUC} = \text{AUC}_{\text{clean}} - \text{AUC}_{\text{adv}}$ aiming to quantify the overall degradation in detector reliability.

4.1.3 Perceptual quality metrics. We assess visual fidelity between the adversarial and original fake images using Structural Similarity Index Measure (SSIM) [32], Peak Signal-to-Noise Ratio (PSNR) [23], LPIPS [36], and the FID metric [8]. Following the AADD 2026, we also consider the aggregate *Challenge Score S*, defined as:

$$S = \sum_{c \in C} \sum_{k=1}^{N} \left(\alpha \cdot \text{SSIM}(I_k, I_k^{ADV}) \right) + \left((1-\alpha) \cdot \left(1 - \text{LPIPS}(I_k, I_k^{ADV})\right) \right) \cdot \left[C_f(I_k^{ADV}) = \text{LABEL}_{\text{real}} \right] \tag{1}$$

where C is the set of all classifiers, N is the number of deepfake images in the test dataset, I_k is the k-th image from the deepfake test set, I_k^{ADV} is the adversarial image generated from I_k, $\text{LABEL}_{\text{real}}$ denotes the real class label, $\alpha = 0.5$, and $[\cdot]$ is the indicator function that is equal to 1 when the predicate holds, and 0 otherwise.

4.1.4 Supplementary VLM-based semantic audit. The AADD metrics quantify detector evasion and perceptual similarity, but they do not directly record whether a visible attribute becomes ambiguous after reconstruction. We therefore conduct a supplementary semantic audit on every original/attacked pair. The audit does not enter the Challenge Score and does not alter DÆLTA or any detector. We use Qwen2.5-VL-3B-Instruct [30], without fine-tuning, as a zero-shot annotator. Each image is processed independently with the same neutral prompt, which requests one short English caption and a JSON record containing `visible_smile`, `eyeglasses`, `head_pose`, `headwear`, and `background_type`. The closed vocabularies include an explicit `unknown` value for an unclear observation.

Let $a_{i,k}^o$ and $a_{i,k}^a$ denote the original and attacked values, respectively, for pair i and attribute k. We use only attributes that are clear in the original, $K_i = \{k : a_{i,k}^o \neq \texttt{unknown}\}$, and define

$$\text{SC}_i = \frac{1}{|K_i|} \sum_{k \in K_i} \mathbf{1}\left[a_{i,k}^o = a_{i,k}^a \right]. \tag{2}$$

Pairs with fewer than three clear original attributes are excluded from the mean, while an attacked `unknown` is counted as a mismatch. We call the resulting mean *Semantic Consistency* (SC). The prompt also constrains captions to directly visible, non-sensitive facts and requires the prefix "A person"; a single correction attempt is permitted for an invalid caption, and deterministic fallback captions are excluded from text metrics.

We report two complementary caption diagnostics. BERTScore F1 compares the attacked caption (candidate) to the original caption (reference) through contextual token embeddings from `roberta-large` [18, 37]; it captures linguistic similarity but does not inspect images. Concept Retention is the directional literal recall of relevant original caption tokens and adjacent bigrams in the attacked caption. Thus, it deliberately penalizes omitted colors or background words even when BERTScore recognizes a paraphrase. Generation is greedy (`do_sample = false`, one beam); model revision, prompt version, software versions, and generation parameters are stored with the results.

4.1.5 Attack Settings. We evaluate DÆLTA against both white-box and black-box detectors. For gradient-based baselines, we implement PGD [20], Auto-PGD (APGD) [5], and MI-FGSM [6] as targeted attacks toward the *real* class with an ℓ_∞ budget of $\epsilon = 8/255$. PGD uses 60 iterations with step size $1/255$, APGD uses 50 iterations with adaptive step size, and MI-FGSM uses 10 iterations with step size $1/255$ and momentum $\mu = 1.0$. All baselines are optimized against an ensemble of ViT-B/16 and DenseNet121-DCT.

For DÆLTA, we adopt the dual-branch autoencoder trained on the 14K FFHQ subset with loss weights $(0.30, 0.60, 0.10)$. Adversarial images are generated via a single forward pass and delta injection, without iterative optimization. ViT-B/16 inputs are processed at 224×224 with ImageNet normalization, while DenseNet121-DCT operates on 128×128 log-scaled DCT representations.

4.2 Ablation Study

4.2.1 Loss Weights Evaluation. To investigate the optimal weight values for each metric in the loss, we analyze the effect of the composite loss-weighting strategy used to train DÆLTA. As defined

above, λ_1 weights the structural term $(1 - \text{SSIM})$, λ_2 weights the perceptual LPIPS term, and λ_3 weights the pixel-wise MSE term. We evaluate different configurations for these parameters in order to find the configuration that maximizes the score S. All configurations are trained and evaluated on the FFHQ dataset using the same fake images and the original AADD evaluation protocol.

Table 1 left side, shows the overall results. As we can see, the configuration $(\lambda_1, \lambda_2, \lambda_3) = (0.30, 0.60, 0.10)$, which assigns the largest weight to LPIPS, achieves the highest Challenge Score of 2003.4075 and the strongest attack performance on both target detectors, with an ASR of 96.12% and 93.44% on ViT-B/16 and DenseNet121-DCT, respectively. The other configurations achieve lower ASR values, ranging from 69.81% to 80.00% on ViT-B/16 and from 37.69% to 49.63% on DenseNet121-DCT, indicating a substantially weaker evasion capability. This degradation is particularly evident for the balanced configuration $(0.33, 0.33, 0.34)$, which yields the lowest ASR on ViT-B/16 (69.81%) among the evaluated settings, suggesting that an even weighting does not translate into a favorable trade-off for either visual quality or attack effectiveness.

Concerning the results for SSIM and LPIPS, the configuration of $\lambda_1 = 0.30$, $\lambda_2 = 0.60$, and $\lambda_3 = 0.10$ achieves values that are slightly lower than those obtained by the alternative weightings, with an SSIM of 0.7513 and an LPIPS of 0.3906, compared to SSIM values ranging from 0.8083 to 0.8113 and LPIPS values ranging from 0.2995 to 0.3138 achieved by the remaining configurations. Nevertheless, this configuration achieves the highest S and ASR values. While assigning a larger weight to LPIPS in the loss function leads to a moderate deterioration in perceptual quality metrics, it significantly improves attack success, yielding ASR gains of up to 26 and 56 percentage points on ViT-B/16 and DenseNet121-DCT, respectively. As a result, a larger number of adversarial images elude the detectors, thus compensating for the reduction in perceptual fidelity, leading to the higher overall S value.

4.2.2 Architecture and Training-set Size. To examine the contribution of the two encoding branches, we compare three autoencoder variants: Pixel-Æ, which operates only on the RGB image; DCT-Æ, which operates on the log-scaled DCT representation; and the complete dual-branch architecture described in Sec. 3. Each variant is trained on FFHQ subsets of increasing size, corresponding to the 5th, 10th, and 21st percentiles of the distance distribution (approximately 5K, 10K, and 14K images, respectively), as well as on the full FFHQ dataset. All configurations are evaluated on the same fake images, and the results appear on the right in Table 1.

Pixel-only branch. The Pixel-Æ variant is particularly effective against ViT-B/16, reaching an ASR of 95.56% when trained on the 5K subset. Its performance against DenseNet121-DCT is lower, with an ASR of 65.50%. This discrepancy is consistent with the RGB branch acting directly on spatial image characteristics while having a more limited influence on frequency-domain traces. Increasing the training-set size progressively improves reconstruction quality: with the full FFHQ dataset, SSIM increases from 0.8005 ± 0.0731 to 0.8604 ± 0.0592, while LPIPS decreases from 0.3493 ± 0.0815 to 0.2269 ± 0.0636. However, attack effectiveness drops substantially, reaching only 49.69% on ViT-B/16 and 12.62% on DenseNet121-DCT.

Frequency-only branch. The DCT-Æ variant shows a different behavior. The 10K configuration achieves an ASR of 92.38%

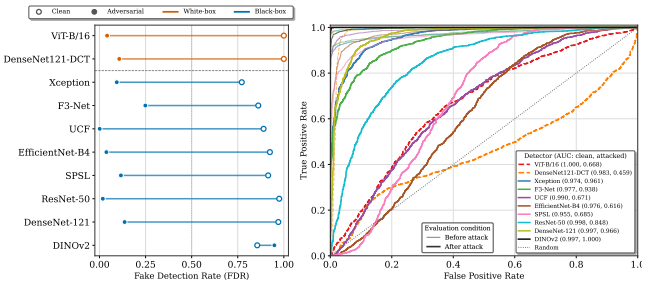


Figure 2: ROC curves and FDR comparison across white-box and black-box detectors before and after the attack.

on ViT-B/16 and 91.88% on DenseNet121-DCT, indicating that the frequency branch can effectively interfere with detectors operating on different input representations. However, this comes at a considerable cost in visual quality: for the same configuration, SSIM is 0.6164 ± 0.1023 and LPIPS is 0.6358 ± 0.0898. Similar values are observed for the other training-set sizes. The DCT branch is therefore effective for detector evasion, but produces overly aggressive modifications when used in isolation.

Dual-branch architecture. The best result is obtained with the complete dual-branch architecture trained on the 14K configuration, corresponding to the 21st percentile. This setting achieves the highest Challenge Score in the table, $S = 2003.4075$, together with an ASR of 96.12% on ViT-B/16 and 93.44% on DenseNet121-DCT. The corresponding SSIM and LPIPS values are 0.7513 ± 0.0825 and 0.3906 ± 0.0790, respectively. Moreover, the advantage of the dual-branch architecture becomes clearer when the models trained on the same subset are compared directly. In the 14K setting, Pixel-Æ achieves ASR values of 78.44% and 27.12%, while DCT-Æ reaches 82.81% and 68.38%. Combining the two branches increases these values to 96.12% and 93.44%. The improvement cannot be attributed to either representation alone: the RGB branch captures spatial structures, the DCT branch captures frequency components, and their latent fusion lets the model exploit both.

Effect of training-set size. It is worth noting that attack effectiveness does not increase monotonically with the amount of training data. When trained on the complete FFHQ dataset, the dual-branch architecture achieves better perceptual quality, with an SSIM of 0.8032 ± 0.0756 and an LPIPS of 0.2860 ± 0.0709, but its ASR decreases to 36.56% on ViT-B/16 and 40.62% on DenseNet121-DCT. This suggests that a broader training distribution makes the autoencoder better at reconstructing a wide range of inputs, but less effective at imposing the compact real-face prior on which the attack relies. The subset selected at the 21st percentile provides the most favorable balance between real-image variability, attack effectiveness, and visual quality. We therefore adopt the DÆLTA-14K setting for the main experiments.

4.2.3 Training Dataset Ablation. After selecting the dual-branch architecture and the 14K subset size (corresponding to the 21st percentile of FFHQ), we investigate how the source of the real

$S_{\text{White-box}}$	$S_{\text{Black-box}}$	SSIM	PSNR	LPIPS	FID
2003.4075	6970.3003	$0.7513_{\pm 0.0825}$	$15.97_{\pm 2.14}$ dB	$0.3906_{\pm 0.0790}$	186.49

Table 2: Results of DÆLTA on the AADD Challenge 2026 test set considering 2 white-box and 8 black-box classifiers.

Training Dataset	S	SSIM	LPIPS	ASR (%)	
				ViT-B/16	DenseNet-DCT
FFHQ [11]	**2003.4075**	$\mathbf{0.7513}_{\pm 0.0825}$	$\mathbf{0.3906}_{\pm 0.0790}$	**96.12**	**93.44**
VGGFace2 [2]	1075.4146	$0.7786_{\pm 0.0784}$	$0.3429_{\pm 0.0810}$	68.56	25.44
CelebA [19]	1628.3008	$0.7763_{\pm 0.0780}$	$0.3500_{\pm 0.0806}$	86.56	56.75
MIX	705.6902	$0.8012_{\pm 0.0772}$	$0.2820_{\pm 0.0756}$	32.25	26.56

Table 3: Comparative evaluation of DÆLTA trained on different datasets.

training images affects the performance of DÆLTA. The same architecture is trained separately on FFHQ, VGGFace2, CelebA, and a mixed dataset, denoted as MIX, obtained by combining the three sources. The filtering procedure described in Sec. 3.1 is applied independently to each dataset, while the architecture, loss function, and evaluation protocol are kept unchanged. All configurations are then tested on the same set of fake images.

As reported in Table 3, FFHQ provides the strongest overall result, achieving a Challenge Score of $S = 2003.4075$. The corresponding ASR is 96.12% on ViT-B/16 and 93.44% on DenseNet121-DCT. CelebA is the second most effective training source, with a score of 1628.3008 and ASR values of 86.56% and 56.75%, respectively. Training on VGGFace2 retains a moderate attack effectiveness against ViT-B/16, with an ASR of 68.56%, but performs considerably worse against DenseNet121-DCT, for which the ASR drops to 25.44%. The influence of the training dataset is particularly evident for the frequency-based detector. The ASR on DenseNet121-DCT decreases from 93.44% with FFHQ to 56.75% with CelebA and 25.44% with VGGFace2. One possible interpretation is that the reconstruction prior learned from FFHQ is more compatible with the real-image characteristics acquired by the two detectors during training. In other words, the representations produced by the FFHQ-trained autoencoder may lie in a region of the feature space that the classifiers more readily associate with the real class.

The MIX configuration shows a different trade-off. Despite achieving the best perceptual quality, with an SSIM of about 0.8012 and an LPIPS of about 0.2820, it obtains the lowest score, equal to 705.6902. Its ASR decreases to 32.25% on ViT-B/16 and 26.56% on DenseNet121-DCT. Thus, broadening the training distribution improves reconstruction fidelity but weakens the autoencoder's ability to suppress or modify the discriminative cues used by detectors. This can be attributed to the fact that the model learns a less restrictive prior that is better adapted to reconstructing a wide variety of facial images, yet less effective at enforcing the compact real-face manifold needed for successful detector evasion. These results confirm that the real-image distribution used for training the autoencoder directly influences the attack's ability to evade forensic detectors. FFHQ provides the most favorable balance, with the highest ASR on both ViT-B/16 and DenseNet121-DCT, while maintaining competitive perceptual quality. We therefore select the filtered FFHQ subset for the final DÆLTA configuration.

4.3 Attack on White-box Detectors

We evaluate DÆLTA against two white-box detectors from the AADD challenge, ViT-B/16 and DenseNet121-DCT, with results in Table 4. On 1,600 fake test images, DÆLTA attains ASRs of 96.12% against ViT-B/16 and 93.44% against DenseNet121-DCT, indicating that most adversarial images are misclassified as real and that DÆLTA reliably evades state-of-the-art deepfake detectors under a white-box threat model.

The attack effectiveness is also confirmed by the FDR. As reported in Figure 2, for ViT-B/16 the FDR decreases from 1.00 on clean fake images to 0.0413 on the corresponding adversarial images, yielding an absolute reduction of 0.9588. Similarly, for DenseNet121-DCT, the FDR drops from 1.00 to 0.1075, corresponding to a reduction of 0.8925. These results demonstrate that only a small proportion of adversarial samples are still correctly identified as fake, confirming a severe degradation in detection performance.

Concerning the ROC-AUC values, as we can see, the AUC decreases from 0.9995 to 0.6675 for ViT-B/16, and for DenseNet121-DCT shows a larger reduction of 0.5241, i.e., from 0.9832 to 0.4591. The post-attack AUC of DenseNet121-DCT falls below the random-guessing threshold of 0.5, suggesting that the attack affects the detector's ability to discriminate between real and fake images.

Regarding visual quality, Table 2 shows that DÆLTA preserves a moderate degree of perceptual fidelity to the original fake images, with an SSIM of 0.7513 and a PSNR of 15.97 dB. The PSNR value combined with the LPIPS of 0.3906 reflects the trade-off discussed in Sec. 3: the DCT branch introduces pixel-wise deviations from the original image in order to attenuate spectral fingerprints, which is reflected in a comparatively high FID of 186.49. Nevertheless, the resulting score S of 2003.4075 indicates that the loss tends to optimize the pixel-level fidelity, as captured by the metric S.

4.4 Attack on Black-box Detectors

To assess the transferability of the proposed approach, we evaluate the generated adversarial examples against a set of unseen black-box detectors with heterogeneous architectures, i.e., ResNet50 [1], DenseNet121 [1], DINOv2 [21], F3Net [24], Xception [26], UCF [34], EfficientNet-B4 [29], and SPSL [17]. Table 4 shows the achieved results, with an overall challenge score S of 6970.30 obtained by DÆLTA across the considered black-box detectors. As we can see, DÆLTA exhibits strong transferability across most detectors, achieving ASR values above 85% on six out of eight black-box detectors, i.e., UCF (99.88%), ResNet50 (98.19%), EfficientNet-B4 (96.25%), Xception (90.62%), SPSL (88.38%), and DenseNet121 (86.38%). Against F3Net, DÆLTA achieves lower performances, with an ASR of 75.06%. Nevertheless, this shows that the learned adversarial patterns generalize across heterogeneous detector architectures.

Concerning the FDR, Fig. 2 provides a direct comparison between detector performance before and after the attack. The hollow dots ○ indicate the FDR on clean fake images, while the other types ● indicate the FDR after the attack. Together, they show how effectively the attack causes fake images to be classified as real. The largest reductions in FDR are observed for ResNet50, from 0.9750 to 0.0181, UCF, from 0.8900 to 0.0013, and EfficientNet-B4, from 0.9244 to 0.0375, while slightly smaller reductions are observed for Xception, F3Net, SPSL, and DenseNet121. This confirms the transferability of the attack to models with heterogeneous architectures.

The ROC curves in Fig. 2 further illustrate the effect of the attack over the full range of classification thresholds. Each color identifies a detector, with lighter and thinner curves representing performance before the attack and darker and thicker curves representing performance after the attack. Dashed curves correspond to white-box detectors, whereas solid curves correspond to

		PGD [20]	APGD [5]	MI-FGSM [6]	**DÆLTA**
White-box	ViT-B/16	95.88	**96.94**	25.25	96.12
	DenseNet121-DCT	57.94	18.50	10.88	**93.44**
Black-box	ResNet50 [1]	5.25	5.94	6.62	**98.19**
	DenseNet121 [1]	6.00	6.81	7.94	**86.38**
	DINOv2 [21]	**9.44**	9.00	9.06	5.13
	F3Net [24]	9.25	10.75	16.25	**75.06**
	Xception [26]	18.94	19.25	25.56	**90.62**
	UCF [34]	15.44	16.88	16.12	**99.88**
	EfficientNet-B4 [29]	11.19	11.75	12.12	**96.25**
	SPSL [17]	16.19	17.25	19.69	**88.38**

Table 4: ASR(%) comparison with existing approaches.

black-box detectors. The largest black-box AUC degradations are observed for EfficientNet-B4, from 0.9764 to 0.6164; UCF, from 0.9900 to 0.6713; and SPSL, from 0.9551 to 0.6847. Conversely, Xception, F3Net, and DenseNet121 retain relatively high post-attack AUC values of 0.9614, 0.9383, and 0.9657, respectively, despite their high ASRs and substantial FDR reductions. This suggests that images generated by DÆLTA are classified as real, yielding the classification score under the decision thresholds of detectors.

DINOv2 is the main exception with an ASR of 3.73%, while its FDR increases from 0.8556 to 0.9488. Its AUC also increases slightly, from 0.9974 to approximately 1.0000. The nearly overlapping clean and adversarial ROC curves confirm its robustness to the transferred perturbations. However, the overall results indicate that the adversarial patterns learned against the white-box models generalize across heterogeneous detector architectures.

4.5 Comparison with Existing Attacks

To assess the effectiveness of DÆLTA, we compare it against three established gradient-based attacks, i.e., PGD, APGD, and MI-FGSM, against white- and black-box detectors. The configuration settings of these approaches have been detailed in Sec. 4. To compare these approaches, we applied them to the 1,600 fake images provided by the challenge organizers and evaluated using the same protocol adopted for DÆLTA. The baselines were configured to induce a real-class prediction and optimized against an ensemble of ViT-B/16 and DenseNet121-DCT. The resulting adversarial images were then evaluated on the black-box detectors without further optimization or access to their parameters, thus measuring cross-detector transferability. Table 4 shows the resulting ASR for both types of models and for each gradient-based attack. As we can see, DÆLTA achieves good ASR performances across most black-box detectors. In particular, it obtains ASR values of 98.19% on ResNet50, 86.38% on DenseNet121, 90.62% on Xception, 99.88% on UCF, and 96.25% on EfficientNet-B4. These results substantially outperform the iterative gradient-based baselines on nearly all black-box detectors, except for DINOv2, where PGD achieves the highest ASR (9.44%), while DÆLTA reaches 5.13%. Nevertheless, it is important to note that all methods exhibit very low ASR values on this detector, with results remaining below 10%. Therefore, these results show that DINOv2 represents a particularly challenging detector for all considered attacks. Overall, these results indicate that the reconstruction-based perturbations produced by DÆLTA transfer effectively across detectors with substantially different architectures and training conditions, supporting the idea that the proposed approach does not depend on a specific detector.

4.6 Supplementary Semantic Audit

We apply the protocol in Sec. 4.1.4 to all 1,600 AADD fake/attacked pairs. The final run contains no annotation errors. Sixty-six 1792×1792 pairs that initially exceeded GPU memory were reprocessed symmetrically with a maximum VLM input side of 1024 pixels, only in memory; the source images were not modified. Table 5 reports the aggregate results. SC is 0.6062 on 1,057 eligible pairs; the remaining 543 pairs have fewer than three clear original attributes and are not failures. The attribute-weighted check is 2,653/4,516 = 0.5875.

Measure	**Value**	**Eligible pairs**
Semantic Consistency	0.6062	1,057 / 1,600
Attribute-weighted retention	0.5875	2,653 / 4,516
BERTScore F1	0.9507	1,508 / 1,600
Concept Retention	0.2612	1,508 / 1,600

Table 5: Aggregate supplementary semantic-audit results. The three metrics have different inputs and are not combined with one another or with the AADD Challenge Score.

The high BERTScore and the lower SC are complementary rather than contradictory. Captions remain linguistically similar, whereas the controlled attributes reveal that pose and scene interpretation are less stable. On the 993 pairs for which all three measures are available, Pearson correlation is only $r = 0.1042$ between SC and BERTScore, and $r = 0.1173$ between SC and Concept Retention. Attacked captions are also shorter on average (8.951 tokens versus 10.355), supporting the interpretation that the VLM often produces a more generic description while retaining the subject.

Attribute	**Clear**	**Match**	**Retention (%)**
Visible smile	675	460	68.1
Eyeglasses	116	111	95.7
Head pose	1,600	922	57.6
Headwear	535	418	78.1
Background type	1,590	742	46.7

Table 6: Per-attribute Semantic Consistency. "Clear" counts the non-`unknown` original values.

Table 6 shows that the signal is not an indiscriminate loss of details: eyeglasses and headwear are generally retained, although the eyeglasses denominator is small. By contrast, head pose and background type have the lowest retention. The most frequent pose transition is frontal to three-quarter (576 cases); for background, outdoor to indoor occurs in 273 cases and the number of `unknown` values increases from 10 to 251. These changes are consistent with a global ambiguity in scene interpretation rather than with complete removal of local subject details.

Repeatability and qualitative observation. Before the full run, a 20-pair pilot obtained SC = 0.5735 on 17 eligible pairs, BERTScore F1 = 0.9570 on 19 valid caption pairs, and Concept Retention = 0.3429. We then repeated greedy inference without cache three times for each of the 40 pilot images (original and attacked), for 120 fresh annotations. The complete five-attribute vector, caption, and JSON output were identical across repetitions (agreement = 1.0). This verifies repeatability of the fixed decoding configuration, not the

correctness of every VLM label. A qualitative visual inspection by the authors additionally identified a semi-transparent double-exposure/face-like overlay in attacked examples: local clothing and facial contours remain visible, while face pose and background become harder to read. This observation is reported only as qualitative evidence, not as a formal human-study score.

Scope and limitations. No directly comparable leaderboard exists for post-hoc semantic preservation between paired original and attacked deepfake portraits. Our five-attribute audit is therefore an interpretable diagnostic, not a claim of state-of-the-art performance. It measures the perception of one VLM with one fixed prompt, covers only selected attributes, and does not establish human ground truth. Question-based faithfulness methods such as TIFA [9] motivate the use of atomic facts, but they address a different text-to-image setting. Future work could compare an independent VLM annotator, probe semantically equivalent prompts, add structured multi-rater human labels, and evaluate complementary image-text grounding measures such as VQAScore [16]; none of these future directions is included in the results above.

5 Conclusion and Future Directions

We presented DÆLTA, a transfer attack that projects synthetic faces toward a compact real-face manifold. Its dual-branch autoencoder jointly processes RGB and DCT information without detector gradients or iterative queries. The results show high ASR and strong transferability across many black-box detectors, outperforming gradient-based baselines, such as PGD, APGD, and MI-FGSM, against multiple detectors. The supplementary semantic audit adds that captions remain highly similar in contextual embedding space (BERTScore F1 = 0.9507), while the controlled Semantic Consistency is lower (0.6062), especially for head pose and background type; this exposes a change in VLM perception that is not captured by the official visual metrics alone.

In the future, we would like to extend DÆLTA to multimodal scenarios where it is necessary to preserve temporal consistency and explore adaptive variants. Moreover, since the challenge score also accounts for robustness to media compression, such as social media, JPEG, and JPEG-AI compressions, we further plan to extend the evaluation of DÆLTA against these transformations in order to better understand the effectiveness of the manifold-projection after such post-processing approaches. The supplementary audit further motivates future semantic validation with independent VLMs and structured human labels, so that the perceptual consequences of manifold projection can be characterized beyond pixel-level similarity without changing the official AADD evaluation.

Reproducibility Statement

To facilitate reproducibility, we release our full implementation, including the dual-branch autoencoder architecture, training configuration, inference-time delta-attack pipeline, and the supplementary semantic-audit code on the official repository[1]. All hyperparameters used to train DÆLTA, including the weights associated with the different loss configurations, network architecture details, and optimizer settings, are detailed in the README. The AADD 2026 dataset used for evaluation will be publicly shared by the challenge organizers.

[1] https://github.com/antonio-Landi/AADD$_C$*hallenge*

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